

TRANSPORT AND THE SUPERCRITICALITY GAP IN 3D NAVIER–STOKES, I

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ABSTRACT. We examine, term by term, the evolution equation of the strain tensor of the three-dimensional incompressible Navier–Stokes equations, $D_t S = -S^2 - \Omega^2 - \nabla^2 p + \nu \Delta S$, and ask which term could carry an estimate closing the gap between the energy level and the critical level. Two elementary lemmas show that on the set where strain produces enstrophy and dominates rotation (in a sense made precise below), every local term, including the centrifugal term, drives the eigenvalues toward the restricted-Euler singular configuration: the most compressive eigenvalue down, the intermediate one up. The anisotropic part of the pressure Hessian is the only inviscid term that can oppose either. At the level of norms the pressure behaves differently. Its contribution to $\frac{d}{dt} \|S\|_{L^{2m}}^{2m}$ vanishes identically at $m = 1$, and its derivative in m at $m = 1$ is a single number Λ that does not depend on the units of $|S|$; whenever $\Lambda \neq 0$ the pressure term changes sign exactly at L^2 . In direct numerical simulations at 128^3 we find $\Lambda < 0$ in every dynamical state: the pressure slows the growth of the strain norms above L^2 and accelerates the growth of the norms below it, including the critical norm $L^{3/2}$, at every exponent tested. This is the norm-level form of a redistribution already observed at high Reynolds number by Buaria, Pumir and Bodenschatz. Advection, finally, contributes nothing to the budget of any Lebesgue norm of any quantity. With every term on the right-hand side accounted for, the regularizing mechanism required by Miller’s model family, if the equations are regular, must act through transport, and no single-time norm budget can see it. Lagrangian tracers show how strain-producing events actually end: each event yields a bounded number of compression e-folds and is terminated without viscosity, by roll-up into vorticity or by pressure-driven flattening. We state what the missing estimate must therefore control: the escalation from one event to the next, not the amplification within one.

1. INTRODUCTION

We write the incompressible Navier–Stokes equations on $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ as

$$\partial_t u - \nu \Delta u + (u \cdot \nabla)u + \nabla p = 0, \quad \nabla \cdot u = 0, \quad (1)$$

with velocity gradient $A = \nabla u$, $A_{ij} = \partial_j u_i$, strain $S = \frac{1}{2}(A + A^T)$, rotation $\Omega = \frac{1}{2}(A - A^T)$, vorticity $\omega = \text{curl } u$, and pressure Hessian $H = \nabla^2 p$. Taking the gradient of (1) and its symmetric part gives

$$D_t S = -S^2 - \Omega^2 - H + \nu \Delta S, \quad D_t = \partial_t + u \cdot \nabla, \quad \Delta p = -\text{tr}(A^2). \quad (2)$$

The only unconditional bound on (1) is the energy inequality, which sits one half of a derivative below the critical level; every known regularity criterion is stated at the critical level [21, 11, 2]. The companion note [24] lists eight conditions that any quantity closing this gap must meet. Its transport condition (C8 there) is read off from a one-parameter family of strain models due to Miller [17]: the family shares the constraint space, the scaling and the enstrophy identity of (1), is regular at one end, and blows up in its interior [16]. If (1) is regular, the regularizing mechanism must therefore involve advection.

This paper makes that condition concrete by eliminating the alternatives. Equation (2) has four terms on its right-hand side and one on its left. We ask of each whether it can carry an estimate at or above the critical level, and we find a reason in every case why it cannot. The reasons are of three kinds: pointwise lemmas (§3), an exact identity together with its first variation (§4), and a kinematic fact about transport (§5). Numerical simulations (§6) confirm each conclusion in real flows and, through Lagrangian tracers, show how strain-producing events end. Section 7 states what the missing estimate must control.

What is new, and what is not. The phenomenology of §4, that the pressure Hessian depletes intense strain and amplifies weak strain, and that its unconditional mean vanishes, is due to Buaria, Pumir and Bodenschatz [7], who observed it at Taylor-scale Reynolds numbers up to 1300; see also [6]. That the anisotropic pressure Hessian opposes the restricted-Euler dynamics is classical [23, 8, 19, 18, 14]. That advection drops out of L^p estimates is textbook. That any regularity proof must use structure of the nonlinearity beyond the energy identity and harmonic-analysis bounds is Tao's conclusion [22]; that it must involve advection is Miller's [17]. What we add is:

- the two lemmas of §3, which we believe are folklore but have not found stated;
- the translation of the pressure's redistribution into a statement about strain norms: a sign change located exactly at L^2 by an identity, with the critical norm on the unfavourable side (§4);
- the resulting term-by-term elimination (Table 1);
- a Lagrangian attribution of how strain-producing events end (§6.5), and the specification of §7.

None of this closes the gap. Its purpose is to say precisely where the missing estimate has to live.

2. THE EIGENVALUE BUDGET

Let $\lambda_1 \leq \lambda_2 \leq \lambda_3$ be the eigenvalues of S , with $\lambda_1 + \lambda_2 + \lambda_3 = 0$, and let e_1, e_2, e_3 be orthonormal eigenvectors. Define

$$P = -4 \det S = -4\lambda_1\lambda_2\lambda_3, \quad Q_A = \operatorname{tr}(A^2) = |S|^2 - \frac{1}{2}|\omega|^2.$$

The quantity P is the local rate of strain self-amplification: $\int P dx = \int \omega \cdot S\omega dx$ for divergence-free u [3]. We call $\{P > 0\}$ the *producing set* and $\Sigma = \{P > 0, Q_A > 0\}$ its *strain-dominated* part. On $\{P > 0\}$ one has $\lambda_1 < 0 < \lambda_2 \leq \lambda_3$: indeed $\lambda_1 < 0 < \lambda_3$ whenever $S \neq 0$, so $P > 0$ forces $\lambda_2 > 0$. In particular λ_1 is a simple eigenvalue there.

Since $\Omega v = \frac{1}{2}\omega \times v$, one has $-\Omega^2 = \frac{1}{4}(|\omega|^2 I - \omega \otimes \omega)$. Write $H_{\text{dev}} = H - \frac{1}{3}(\operatorname{tr} H)I$ and note $\operatorname{tr} H = \Delta p = -Q_A$. Where λ_i is simple, the Hellmann–Feynman formula $D_t \lambda_i = e_i \cdot (D_t S) e_i$ applied to (2) gives the exact budget

$$D_t \lambda_i = \underbrace{-\lambda_i^2 + \frac{1}{3}Q_A}_{L_i} + \underbrace{\frac{1}{4}(|\omega|^2 - (\omega \cdot e_i)^2)}_{V_i} - \underbrace{e_i \cdot H_{\text{dev}} e_i}_{\Pi_i} + \underbrace{\nu e_i \cdot \Delta S e_i}_{N_i}. \quad (3)$$

The term L_i is the restricted-Euler part: the local strain and the isotropic part of the pressure, which is itself local since $\operatorname{tr} H = -Q_A$. The term $V_i \geq 0$ is centrifugal. The anisotropic pressure Hessian Π_i is the only nonlocal term, and N_i is viscous. Dropping Π_i and N_i gives the restricted Euler system of Vieillefosse [23], which blows up in finite time for generic data, along the configuration $\lambda \propto (-2, 1, 1)$ [8].

Remark 2.1 (a one-number closure). In terms of the normal share $\eta = e_1 \cdot H e_1 / \operatorname{tr} H$ of the pressure Hessian, $D_t \lambda_1 = -\lambda_1^2 + \eta Q_A + V_1 + N_1$. For pure strain along $\lambda = (-2, 1, 1)\mu$ this

reads $D_t\mu = (2 - 3\eta)\mu^2$, so the restricted-Euler blowup is arrested exactly when $\eta > 2/3$. For an ideal flat sheet, whose source depends on one coordinate only, $\eta = 1$; for a uniform oblate ellipsoid η is the normal depolarisation factor, which equals $2/3$ at aspect ratio 3.41. This suggests that flattening arms its own brake. Section 6.2 shows that this pointwise picture does *not* describe real flows.

3. THE LOCAL TERMS DRIVE THE SINGULAR CONFIGURATION

Lemma 3.1. *On $\Sigma = \{P > 0, Q_A > 0\}$ one has $L_1 + V_1 < 0$.*

Proof. On $\{P > 0\}$, $\lambda_2\lambda_3 > 0$, so $\lambda_1^2 = (\lambda_2 + \lambda_3)^2 \geq \lambda_2^2 + \lambda_3^2$ and hence $2\lambda_1^2 \geq |S|^2$. The condition $Q_A > 0$ means $|\omega|^2 < 2|S|^2$. Using $Q_A = |S|^2 - \frac{1}{2}|\omega|^2$,

$$L_1 + V_1 = -\lambda_1^2 + \frac{1}{3}|S|^2 + \frac{1}{12}|\omega|^2 - \frac{1}{4}(\omega \cdot e_1)^2 < -\frac{1}{2}|S|^2 + \frac{1}{3}|S|^2 + \frac{1}{6}|S|^2 = 0. \quad \square$$

Lemma 3.2. *On $\{P > 0, |\omega| < |S|\}$ one has $L_2 + V_2 > 0$.*

Proof. On $\{P > 0\}$, $0 < \lambda_2 \leq \lambda_3$ and $\lambda_1 = -(\lambda_2 + \lambda_3)$, so $|S|^2 \geq \lambda_2^2 + \lambda_2^2 + 4\lambda_2^2 = 6\lambda_2^2$. Using $(\omega \cdot e_2)^2 \leq |\omega|^2$,

$$L_2 + V_2 = -\lambda_2^2 + \frac{1}{3}|S|^2 + \frac{1}{12}|\omega|^2 - \frac{1}{4}(\omega \cdot e_2)^2 \geq -\frac{1}{6}|S|^2 + \frac{1}{3}|S|^2 - \frac{1}{6}|\omega|^2 = \frac{1}{6}(|S|^2 - |\omega|^2) > 0. \quad \square$$

Corollary 3.3. *On the strain-dominated producing set, the local inviscid terms push λ_1 down and, where $|\omega| < |S|$, push λ_2 up, that is, toward the restricted-Euler configuration and toward the planar stretching that the middle-eigenvalue criterion [15] identifies as necessary for blowup. The anisotropic pressure Hessian is the only inviscid term in (3) that can oppose either motion.*

Both inequalities are sharp at the boundary of their sets (for instance at $\lambda_2 \rightarrow 0$, $|\omega|^2 \rightarrow 2|S|^2$ in Lemma 3.1); a check on 1.4×10^6 random traceless matrices in Σ gave $\max(L_1 + V_1)/\lambda_1^2 = -0.0055$. The centrifugal term helps, but it can never reverse the local drive on these sets. Lemma 3.1 is the pointwise sense in which the pressure is indispensable; §4 shows that at the level of norms it cannot be sufficient.

4. THE PRESSURE IS INVISIBLE AT L^2

Proposition 4.1. *For smooth divergence-free u on $\mathbb{T}^3$, $\int H : S \, dx = 0$.*

Proof. Integrating by parts twice, $\int \partial_i \partial_j p \, S_{ij} = \int p \, \partial_j \partial_j S_{ij}$, and $\partial_j S_{ij} = \frac{1}{2}(\Delta u_i + \partial_i \nabla \cdot u) = \frac{1}{2}\Delta u_i$, whose divergence vanishes. $\square$

For $m > 0$ set $w_m = |S|^{2m-2}$ and, along solutions of (1),

$$\begin{aligned} D_{ss}(m) &= - \int w_m \operatorname{tr} S^3, & D_{vs}(m) &= -\frac{1}{4} \int w_m \omega \cdot S \omega, \\ \Pi(m) &= - \int w_m H : S, & N(m) &= \nu \int w_m S : \Delta S. \end{aligned}$$

For $m < 1$ we assume S does not vanish, which is generic in three dimensions since S has five independent components; otherwise one regularizes $|S|^2 \mapsto |S|^2 + \varepsilon$.

Proposition 4.2 (the L^{2m} strain budget). *For smooth solutions,*

$$\frac{d}{dt} \frac{1}{2m} \int |S|^{2m} \, dx = D_{ss}(m) + D_{vs}(m) + \Pi(m) + N(m).$$

At $m = 1$: $\Pi(1) = 0$ and $D_{vs}(1) = -\frac{1}{3}D_{ss}(1)$, so that $\frac{d}{dt} \|S\|_{L^2}^2 = -4 \int \det S \, dx + 2N(1)$.

Proof. Contract (2) with $w_m S$. One has $S : (-S^2) = -\operatorname{tr} S^3$, $S : (-\Omega^2) = \frac{1}{4}(|\omega|^2 \operatorname{tr} S - \omega \cdot S \omega) = -\frac{1}{4}\omega \cdot S \omega$, and $w_m S : D_t S = D_t(|S|^{2m}/2m)$. The advective part integrates to zero since $\nabla \cdot u = 0$. At $m = 1$, $\Pi(1) = 0$ is Proposition 4.1; $\operatorname{tr} S^3 = 3 \det S$ for traceless S , and $\int \omega \cdot S \omega = -4 \int \det S$ [3], which give the ratio $-1/3$ and the identity of Neustupa–Penel and Miller [15]. $\square$

At the enstrophy level every inviscid term is thus fixed by an identity: vorticity removes exactly one third of the self-amplification, the pressure removes exactly nothing, and the whole difficulty is concentrated in $-\int \det S$. Away from $m = 1$ the pressure is present, and its behaviour near $m = 1$ is governed by one number.

Definition 4.3. For S nonvanishing, let $\Lambda = \frac{d\Pi}{dm}\Big|_{m=1} = - \int \log |S|^2 H : S \, dx$.

By Proposition 4.1, Λ is unchanged if $\log |S|^2$ is replaced by $\log(c|S|^2)$ for any constant $c > 0$: it does not depend on the units in which the strain is measured. It scales like $\frac{d}{dt}\|S\|_{L^2}^2$.

Corollary 4.4 (the sign switch). *If $\Lambda \neq 0$, there is $\delta > 0$ such that for $0 < |m - 1| < \delta$ the sign of $\Pi(m)$ is the sign of $\Lambda(m - 1)$. In particular, if $\Lambda < 0$ the pressure decreases the growth rate of $\|S\|_{L^{2m}}$ for m slightly above 1 and increases it for m slightly below 1.*

Proof. $m \mapsto \Pi(m)$ is smooth, $\Pi(1) = 0$ and $\Pi'(1) = \Lambda$. $\square$

Remark 4.5 (where the critical norm sits). The norm $\|S\|_{L^q}$, equivalently $\|\nabla u\|_{L^q}$ for $1 < q < \infty$, is critical for $q = 3/2$, that is $m = 3/4 < 1$. When $\Lambda < 0$ and the sign pattern of Corollary 4.4 persists down to $m = 3/4$, which is what we observe in every dynamical state (§6.4), the pressure accelerates the growth of the critical norm and of the subcritical window $3/2 < q < 2$, and helps only above L^2 . The location of the switch is not a property of the flow: it is fixed at L^2 by Proposition 4.1, the one level at which the pressure cannot act. The redistribution itself is the phenomenon of [7]: conditioned on the strain amplitude, the pressure Hessian amplifies weak strain and depletes intense strain, with the crossover near the mean amplitude. Corollary 4.4 is its translation into norms.

5. TRANSPORT IS INVISIBLE TO EVERY LEBESGUE BUDGET

Proposition 5.1. *Let u be smooth and divergence-free on $\mathbb{T}^3$, and let F be any C^1 function of the velocity gradient. Then $\int u \cdot \nabla F(A) \, dx = 0$. Consequently, in any identity of the form $\frac{d}{dt} \int F(A) \, dx = \int F'(A) : (D_t A) \, dx$, the advection term contributes nothing.*

The proof is one integration by parts. The consequence is less trivial than the proof. The pressure is invisible at one exponent; advection is invisible at all of them, and for every functional of the gradient that is an integral of a pointwise function. Its effect enters such budgets only indirectly, through the dependence of the nonlocal terms H and, via Biot–Savart, ω on where the fluid has been carried. An estimate built from single-time integral functionals cannot credit transport with anything.

Remark 5.2. Table 1 is consistent with the two known obstructions. Tao’s averaged equation [22] keeps the energy identity and every harmonic-analysis bound on the nonlinearity, but not, in general, its form as transport by the velocity field, and it blows up. Miller’s family [17] keeps the constraint space, the scaling and the identity of Proposition 4.2 at $m = 1$, varies only the coefficient of the nonlinearity, and blows up at $\mu = 2/3$ [16]; the case $\mu = 1$ is (1) without advection. Both constructions discard the one term that no norm budget registers.

term in (2)	pointwise on Σ	at L^2	at $L^{3/2}$ (critical)
$-S^2$, $-\frac{1}{3}(\text{tr } H)I$ $-\Omega^2$	drives $\lambda_1 \downarrow$, $\lambda_2 \uparrow$ (Lem. 3.1, 3.2) brakes, never reverses (Lem. 3.1)	the whole problem exactly $-\frac{1}{3}$ (Betchov)	drives brakes; ratio < 1 at intense strain [7]
$-H_{\text{dev}}$	only inviscid brake (Cor. 3.3)	exactly 0 (Prop. 4.1)	wrong sign if $\Lambda < 0$
$\nu \Delta S$	small at exits (§6.5)	dissipates	supercritical scaling
$u \cdot \nabla S$	carries the fluid	exactly 0	exactly 0 (Prop. 5.1)

TABLE 1. The ledger. Entries citing a lemma, a proposition or an identity are theorems; the others are observations, from §6 or from the high-Reynolds-number data of [7]. The entry for the pressure at the critical level is conditional on $\Lambda < 0$, which holds in every dynamical state we computed.

6. NUMERICAL EVIDENCE

6.1. Method. We solve (1) on $\mathbb{T}^3$ with 128^3 Fourier modes, the 2/3 dealiasing rule, the rotational form $u \times \omega$ of the nonlinearity, and a fourth-order Runge–Kutta scheme with an integrating factor for viscosity; $\nu = 1/400$, $\Delta t = 0.01$, $0 \leq t \leq 10$. Two initial data of energy $1/8$ are used: the Taylor–Green vortex (**tg**) [5] and a random-phase field with spectrum $\propto k^4 e^{-2(k/3)^2}$ (**rand**). At peak dissipation the Taylor-scale Reynolds number is about 40 and $k_{\text{max}}\eta \approx 1.3$ – 1.5 . We report the fraction of energy in the shell $0.8k_{\text{max}} < |k| \leq k_{\text{max}}$ and treat states with fraction above 10^{-4} as marginally resolved; the fully resolved dynamical snapshots are **tg** at $t = 2, 4$ and **rand** at $t = 10$. These Reynolds numbers are thirty times smaller than those of [7]; the simulations illustrate and test the statements above in real flows, they do not extend them.

Validation. At 32^3 the energy budget $dE/dt = -2\nu Z$, $Z = \frac{1}{2}\langle |\omega|^2 \rangle$, holds to six digits. The production-weighted mean of $\cos^2(\omega, e_2)$ grows from its isotropic value $1/3$ to 0.5 – 0.7 , the known preferential alignment of vorticity with the intermediate eigenvector [1]. The identity (2), tested against the solver with a second-order time difference and compared on the retained modes, has relative residual 1.8×10^{-5} , 4.5×10^{-6} and 1.1×10^{-6} for $\Delta t = 2, 1, 0.5 \times 10^{-3}$. Proposition 4.1 holds to 10^{-17} relative, the ratio $D_{vs}(1)/D_{ss}(1) = -1/3$ to four digits, and Λ is unchanged under $\log |S|^2 \mapsto \log |S|^2 + 5$ to 10^{-15} . Every hypothesis tested below was written down, with its kill criterion, before the corresponding run.

6.2. The pointwise sieve fails. Remark 2.1 suggests that producing regions should sit at $\eta \geq 2/3$. They do not. Weighted by P on Σ , the fraction of production at points with $\eta < 2/3$ is 0.70 – 0.89 throughout the growth phase of **tg** and 0.75 – 0.77 in **rand**; the Gaussian initial field of **rand**, which has no dynamics, gives 0.76 . The registered kill criterion (above $1/2$ in both runs) fired. The fixed threshold $2/3$ is the most stringent case: by (3) the pure-strain threshold is $\lambda_1^2/|S|^2 \in (\frac{1}{2}, \frac{2}{3}]$. But the conclusion does not depend on this: the full budget, below, shows the same thing without a threshold.

6.3. Eigenvalue budgets. Table 2 gives the terms of (3) for λ_1 over the top 1% of production in Σ , weighted by P and normalized by λ_1^2 (which is at least $|S|^2/2$ there).

The pressure brakes throughout the growth phase in both runs. In **tg** it cancels 90% of the drive while the vortex sheets form ($t = 2.25$); when the sheets roll up ($t = 2.5$ – 3) its share collapses and the centrifugal term takes over, and while production is intense the pressure does not regain the lead. In **rand**, $V_1 > \Pi_1$ from $t = 0.75$ on. Of the three registered criteria

run, t	L_1	V_1	Π_1	N_1	total	Π_1 reverses
tg , 1.0	−0.49	0.00	+0.23	0.01	−0.24	0.00
tg , 2.25	−0.51	0.01	+0.45	0.04	−0.005	0.19
tg , 3.0	−0.59	0.14	+0.07	0.05	−0.32	0.00
tg , 4.0	−0.71	0.32	+0.14	0.12	−0.14	0.07
rand , 0.5	−0.62	0.18	+0.21	0.04	−0.19	0.05
rand , 1.0	−0.74	0.37	+0.14	0.11	−0.13	0.03

TABLE 2. The λ_1 budget over the top 1% of production. The last column is the production share at which $L_1 + V_1 + \Pi_1 > 0$; by Lemma 3.1 this is exactly the share at which the pressure reverses the compression.

(pressure brakes; pressure is the dominant inviscid brake; inviscid braking wins on most of the production), the first passed and the other two failed.

For the sheet-forming eigenvalue λ_2 , on the whole of Σ and normalized by $|S|^2$, the local terms drive ($L_2 + V_2 = +0.17$ to $+0.25$) and the pressure opposes: $\Pi_2 = -0.09$ (**tg**, $t = 2$), -0.15 (**tg**, $t = 4$), -0.14 (**rand**, $t = 10$). In every snapshot from $t = 4$ in **tg** and from $t = 2$ in **rand**, Π_2 cancels a stable 73–87% of the local drive (at **tg**, $t = 2$, during sheet formation, 47%), and it exceeds the viscous term by a factor of 2 to 10. On the eigenvalue that the middle-eigenvalue criterion controls, the pressure is the main brake.

6.4. **Norms.** Table 3 gives $\Pi(m)/D_{ss}(m)$ on the resolved snapshots, with $\Lambda/D_{ss}(1)$.

snapshot	$m = 0.6$	0.75	0.9	1	1.5	2	3	$\Lambda/D_{ss}(1)$
tg , $t = 2$	+0.228	+0.137	+0.053	0	−0.222	−0.387	−0.597	−0.51
tg , $t = 4$	+0.171	+0.094	+0.033	0	−0.107	−0.173	−0.286	−0.30
rand , $t = 10$	+0.104	+0.061	+0.023	0	−0.086	−0.141	−0.199	−0.21

TABLE 3. The pressure term in the L^{2m} strain budget, relative to self-amplification. The column $m = 3/4$ is the critical norm $L^{3/2}$.

The same sign pattern holds at every exponent tested in all ten dynamical snapshots ($t = 2, \dots, 10$ in both runs), with $\Lambda/D_{ss}(1)$ between -0.12 and -0.51 . The Gaussian initial field has the opposite pattern ($\Lambda/D_{ss}(1) = +0.49$): the redistribution is produced by the dynamics, not by the kinematics. At $m = 3/2$, that is for $\|S\|_{L^3}$, the inviscid terms together leave $(D_{ss} + D_{vs} + \Pi)/D_{ss} = 0.55$ – 0.65 of the self-amplification in every dynamical snapshot; the pressure removes between 5% and 22%, vorticity about 30%, and viscosity the rest.

6.5. **How producing events end.** We seeded 32^3 tracers, advanced them with trilinear interpolation of the velocity (interpolation error decreasing by 4.01 per halving of the mesh) by Heun’s method aligned with the solver steps, and sampled (3) along them every 0.05, about a tenth of the Kolmogorov time. A *visit* is a maximal time interval spent in Σ . We keep complete visits (entered after $t = 0$, left inside the window), about 5.8×10^4 in **tg** and 7.2×10^4 in **rand**. A visit ends by *roll-up* if Q_A drops through zero (vorticity takes over) and by *flattening* if P does, that is if λ_2 drops through zero. For each visit let $C = \int |\lambda_1| dt$, the number of compression e-folds it delivers.

All three registered criteria passed. Roll-up ends most of the production, narrowly in **tg**. The most intense visits deliver only 1.5–1.8 times the compression of a typical one: intensity does not buy residence. Over the last 0.25 time units before a flattening exit, weighted by P ,

	tg growth	tg resolved	rand growth	rand resolved
ended by roll-up	0.56	0.55	0.70	0.81
ended by flattening	0.41	0.42	0.27	0.18
C , all visits	3.70	2.90	2.46	1.35
C , top 1% of visits	5.61	4.36	4.20	2.44
median duration, all / top 1%	1.40/1.95	1.40/2.33	0.75/1.50	0.50/0.95

TABLE 4. Visits to Σ , weighted by the production $\int P dt$ of each visit. Growth: $t \leq 6.75$ (**tg**), $t \leq 2.25$ (**rand**); resolved: $t \leq 4.5$, $t \leq 1.25$.

the pressure does not merely brake λ_2 but overpowers the local drive: $(L_2 + V_2, \Pi_2, N_2) = (+3.10, -4.65, -0.68)$ in **tg** and $(+1.31, -2.82, -0.26)$ in **rand**. Viscosity is five to forty times smaller than the pressure at these exits, and also small at roll-up exits. Producing events are ended without viscosity: by vorticity, or by the pressure.

Residence times in vortical and strain-dominated regions have been studied before [4]; we are not aware of the attribution of exits to roll-up and pressure-driven flattening, but make no claim of priority for it.

7. WHAT THE MISSING ESTIMATE MUST CONTROL

The picture that emerges is the following. A strain-producing event lasts one to three Kolmogorov times and delivers a bounded number of compression e-folds. It is ended inviscidly. Viscosity does not stop the event; it stops the sequence of events, at the Kolmogorov scale. We have run a single Reynolds number; the data of [7], over $140 \leq R_\lambda \leq 1300$, show that the conditional ratios of self-amplification, vortex stretching and pressure barely depend on it.

By the criterion of Beale, Kato and Majda in the strain form of Ponce [2, 20], a singularity at T^* requires $\int_0^{T^*} \|S\|_{L^\infty} dt = \infty$. In the language of visits, since each visit contributes a bounded number of e-folds, this requires infinitely many visits before T^* , not necessarily along one trajectory, with intensity escalating from one to the next and duration shrinking fast enough for the total time to stay finite. This is the shape of Tao's averaged blowup [22]: bounded amplification per stage, stages accelerating geometrically. A cap on the amplification within one event is therefore not an obstruction to blowup, and an estimate on single events cannot close the gap. What must be controlled is the escalation across events. By §5, what governs that escalation is invisible to single-time norm budgets.

Question 7.1. *Let a visit to Σ at scale ℓ end by roll-up or by flattening. What fraction of its production is handed to structures at scale $\ell/2$ within the next Kolmogorov time, and does that hand-off ratio differ between the two exit channels? Is there a quantity, Lagrangian, built from commutators of advection with the nonlocal operators [13], or integrated in time along trajectories [10], that bounds the product of successive hand-off ratios?*

A hand-off ratio below one on average, uniformly in the Reynolds number, would be the quantitative form of a cascade that cannot accelerate; a ratio that can exceed one along a sequence is the form a blowup would take. The first half of Question 7.1 is measurable with scale-filtered velocity gradients [12, 9]; it is measured in Part III [25], and the escalation question it leads to is made precise in Part IV [26]. We do not know how to approach the second half.

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